

Word search Puzzles

Paul Curzon

Write a program that is capable of solving the following word search. It should use a general search algorithm at its core. It should be able to solve any other word search given too. How might it be able to find the missing extra name. What information would it need to be able to do this? Try and code your solution providing as general a solution as possible.

Famous Computer Scientists Word search

Lots of names are hidden in the grid below. Words can appear horizontally, vertically or diagonally, running backwards or forwards. Different names can cross and overlap.

Find the following famous computer scientist's names in the word search grid. Their first and second names may be in different places. Spaces, accents and hyphens are not included.

ADA LOVELACE, ANITA BORG, BARBARA LISKOV, DANA ULERY, DOROTHY DENNING, FRAN ALLEN, GRACE HOPPER, JEANNETTE WING, KAREN SPARCK-JONES, MARISSA MAYER, MUFFY CALDER, URSULA MARTIN, WENDY HALL

ALAN TURING, CHRIS STRACHEY, EDGAR CODD, EDSGER DIJKSTRA, JOHN VON NEUMANN, MAURICE WILKES, MOHAMMED AL-KHWARIZMI, NIKLAUS WIRTH, PHILIP EMEAGWALI, SERGEY BRIN, TIM BERNERS-LEE, TONY HOARE, VINT CERF

Which extra name appears in the grid: a forgotten genius of Bletchley Park?

```
N K R E C S N I R U T J E A L U S R U
A A E F R E C A C N I U E W V N H O J
M R O M A K B R I T M O R N H F B Y V
H E A D A L O V E L A C E I M R R S P
C N N I W I R T H O O O P K R A I E H
L S I J P W E N D Y H A L L E N N R I
E P T K K K N V O K S I L A Y I F G L
W A A S C H M A U R I S E U A V Z E I
N R G T A W E E L T R U M S M R S Y P
O C G R O B G E F A J E A N N E T T E
D K Y A A N W X G A R D R L A D I M M
R J H G I C O D D A Z S T O S L M O E
O O T W O O E S O H J G I O S A Z H A
G N O N T P I H E D S E N P I C I A G
R E R E H R Y E O B A R B A R A R M W
A S O U H N C I J P D C G K A L A M A
C B D C O I U H E L P M N Y M M W E L
E O S T R A C H E Y D E I C R N H D I
L R P U D E N N I N G E R O O E K B P
E N A W S R E N R E B M U F F Y L R Q
E M V O N N E U M A N N T D A N A U A
```

Computational Thinking Hint

You may be able to use your natural pattern matching abilities to spot words randomly. An algorithmic thinking approach may work better though even for a human. You need an organised way that is guaranteed to find the words. Take the first letter of a word and start at the top left hand corner, scanning along the rows in turn looking for it. When you find it check in all directions for the second letter. If it is not there, then move on. If it is look for the third letter in the same direction.